

Assignment 5: Final Project

6/9/26

CMPM 125

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Reflection:

For this improvement of the project, I worked on creating the title screen, as well as adding audio (footsteps, background music, sound effects), implementing a new enemy asset (specifically modifying the movement script for it to fit into our game and use sound cues to chase rather than visuals), and also doing some post-processing.

I will say that it did take quite a while to get the footsteps to sound right, especially when going from walking to sprinting to crouching. However, it was pretty rewarding to get it right. For my graphics game, I originally focused on particle systems, but I felt that particle systems didn't really fit in with the backroom theme, so I also had to look up some tutorials on how to do post-processing to make everything look kind of granier/ tinted yellow. I've also never done any 3d animations, and although I imported the animations and hazmat assets, it took me some time to figure out how to get the Animator to work. They still look a little bit wonky, but much better than the original version.

This quarter really helped me learn a lot about 3D game development in Unity. I would say that up to this point, all of the games I've made have been 2D, so this was really helpful. I used to think that Unity was a super complicated tool, and although there are still things I don't understand about it, I feel a lot more confident with it now. If I could go back in time and change anything, I would definitely start working on this project earlier, but with my other final assignments and my graduation coming up, I've been quite overwhelmed. However, I'm still proud of what I've learned so far, even if this final game isn't what I imagined it would be.

Raza Khan

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Reflection:

For this project, my main contribution was creating and expanding the level. I helped turn the original stealth game into a Backrooms-style game by adding a second level, more hallways, rooms, props, a ceiling, and Backrooms-inspired textures. I also helped set up the gameplay flow by making the Level 1 exit teleport the player to Level 2, adding a pink key requirement for the final exit, creating UI feedback for the key, and helping with enemy pathing.

One challenge I overcame was working with Unity assets, materials, colliders, triggers, UI, and scripts. Some imported models showed up wrong, so I learned how to fix shader issues. I also learned how to use triggers for teleporting, key collection, and exit conditions.

With more time, I would expand the game by adding more Backrooms levels, each with different layouts, items, enemies, and escape goals. The possibilities are endless because each level could have a new theme or challenge while still keeping the creepy Backrooms feeling. This project helped me learn how to take a simple stealth game and build it into a bigger experience.

Ronitt Katkoria

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Reflection:

I contributed to this project by implementing a hide ability when you go near walls that allows the player to temporarily evade the guard. The hiding near the walls is inspired by the actual backrooms movie, and stays on theme. The ability can be activated by pressing the E key and makes the player invisible to the detection system of the guard for three seconds. I also added UI elements to communicate the mechanic to the player. There is a message at the start of the game explaining the ability and status messages when the ability is activated.

One technical challenge I faced was integrating the hide ability with the existing guard AI system, because I needed to know how the guard detected and tracked the player before modifying the logic so that the player would be ignored when hidden. Another challenge involved managing the UI elements and making sure messages disappeared in a timely manner so it is not annoying for the player. Working within an existing project helped me gain experience reading and modifying code written by teammates rather than building from scratch.

Through the quarter, I learned the importance of communication and source control when working on team projects. Coordinating features between team members requires planning, so conflicts don't happen, especially working with GitHub. I gained a way better understanding of Unity Project structure, and integrating new mechanics into an existing game like I did with this

one. With more time, polishing would have been done to have audio feedback for hiding, and maybe a cooler animation.

Noam Vardi

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Reflection:

For this final project, I focused mostly on atmosphere, player feedback, and audio polish. After the core stealth gameplay was working, I worked on making the level feel more like the backrooms by darkening the environment and adjusting post-processing so everything looked grainier. I also spent time on the key collection flow feedback loop. I replaced the old text-based key UI with a one-slot hotbar at the bottom of the screen, and added a “FIND THE PINK KEY” objective that appears when you enter the second area and stays up until you actually pick up the key. To make the key easier to spot in the darker level, I added both lighting and emissive material to it so it glows pink and stands out from the rest of the environment. It also added to the mysterious vibe of the level which increased the immersion.

I also implemented enemy audio feedback. When a guard detects you, a one-shot alert sound plays, and during the chase a looping static sound gets louder the closer the guard is to you. That took longer than I expected because I had to write a separate script for the proximity audio, figure out how to keep it from conflicting with the detection sound, and make sure it only played after you were actually spotted rather than all the time. Getting the volume to fade smoothly based on distance was slightly tricky because I was working with prefabs that had been customized in the scene already, but it felt really rewarding once it finally worked, because you can feel the guard getting closer even when you are not looking at them.

I feel a lot more confident with Unity now, especially scene setup, post-processing, and working with audio. If I could go back and change anything, I would start working on this project earlier, but with other final assignments and end of year stuff, I have been pretty overwhelmed. Still, I am proud of what I contributed, and even more so about what ended up being more challenging because I feel that I learned more in those parts.

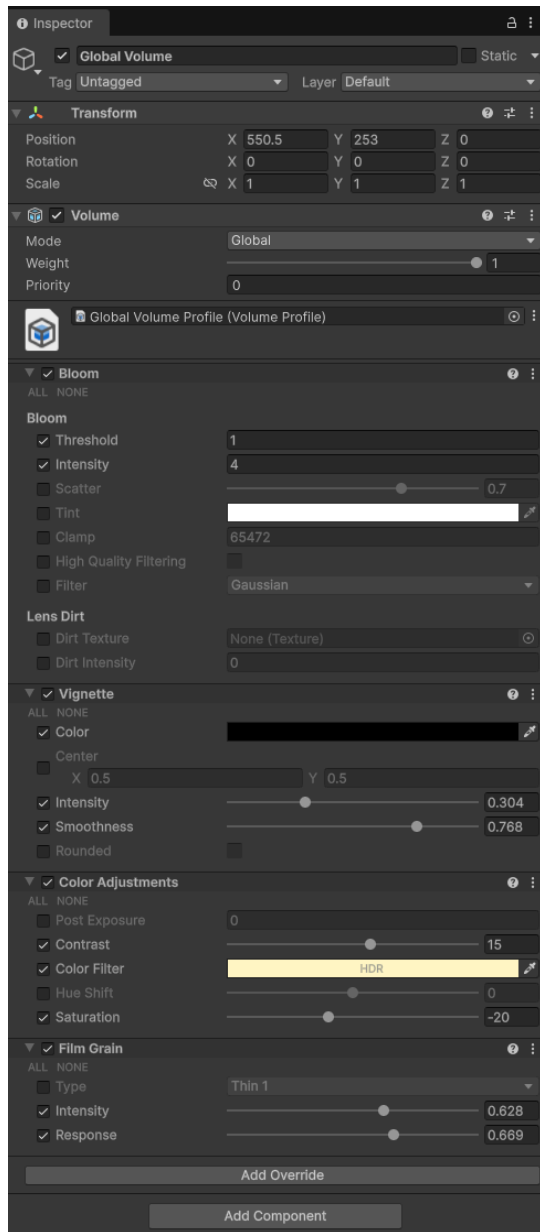
Game Expanded: Stealth Game (using Zosia and Raza’s base game)

Changes:

- Redoing the theme - Backrooms style game

- New enemy - Mimic (chases based on audio cues instead of visuals like the other guards)
- New textures
- Additional levels
- Audio
- Title screen
- Post-processing
- Lighting
- UI and key inventory

Graphics Integration:



Audio Assets:

1. [Entity Aggro Sounds](#)
2. [Background Music](#)
3. [Footstep Clips](#)
4. [Static SFX](#)
5. [Piano Slam](#)

Other Assets:

1. [Guard walking animations](#)
2. [Mimic enemy](#)
3. [Hazmat suit](#)

Screenshot Gallery

